

Viswanathan T J

Software Engineer

✉ viswanathanTJ@gmail.com · 🌐 viswa2k.in · 🐙 github.com/viswanathanTJ · 🌐 linkedin.com/in/viswanathanTJ

SUMMARY

Software Engineer with **3+ years at Zoho**, building production systems in Python and Java — **REST APIs, microservices**, a shared **API gateway**, and **Kafka**-based cross-service data sync. Ships software end to end: design, development, **Docker**, **CI/CD**, and self-hosted operations.

SKILLS

PROGRAMMING	Python	Java	JavaScript	TypeScript	Shell Scripting		
BACKEND	FastAPI	Flask	Django	REST APIs	Microservices	Kafka	SQLAlchemy
FRONTEND	React	Next.js	HTML	CSS			
DATABASES	MySQL	PostgreSQL	Redis	Firebase	MongoDB	SQLite	
CLOUD & DEVOPS	Docker	Google Cloud	GitHub Actions	CI/CD	Linux		
TOOLS & SECURITY	Git	Postman	JWT	RBAC	Cryptography		

EXPERIENCE

Member of Technical Staff

Jun 2023 – Present

Zoho Corporation · Chennai

- Built a CLI-based **server configuration and deployment automation** tool in **Python and Shell**, adopted by the **entire Zoho Projects engineering team** for setup and releases.
- Cut developer cycle time by **~40%** by streamlining environment setup, code changes, and testing loops.
- Developed backend components for a shared **API gateway** powering **install/uninstall for all Zoho Projects integrations** (10+ external systems) — new integrations are **config-driven**, eliminating **~40% of duplicated integration code**.
- Consolidated **real-time data synchronization** from **5+ services** into one shared design that dispatches **Kafka or scheduler payloads** based on each module's sync requirement, cutting cross-service sync latency **from minutes to seconds** — now the standard pattern for new services.

Project Trainee

Aug 2022 – Jun 2023

Zoho Corporation · Chennai

- Independently built a full-stack **project management tool** — **backend APIs and UI** — adopted by **200+ internal users**.
- Resolved **30+ production issues** in a live product and collaborated on **new-feature analysis** with the core team.
- Drove the team's **Eclipse** → **IntelliJ migration** by reimplementing slow IDE-plugin workflows as **Python/Shell scripts** with fully automated configuration.

PROJECTS

Academic Suite — Exam Controller

2025 – Present

FastAPI · React/TypeScript · PostgreSQL · Redis · Pandas · Docker

- Upgraded the **Marksheet Management System** into a VPS-hosted, role-aware web platform — in production since 2025 serving an institution with **200+ staff and 5,000+ students**, architected to scale to **500K+ students**.
- One **FastAPI + PostgreSQL** backend (**Dockerized**, self-deployed with backups and monitoring): staff portal with granular **RBAC** and time-windowed marks entry; **student portal** (tab-scoped **JWT**, AES-GCM sessions) for marksheet and hall-ticket downloads, **reevaluation**, and **photocopy** requests.

Agentic AI Platform with MCP Integration [GitHub](#)

2025

Python · Gemini ADK · MCP · Vertex AI · Cloud Run · A2A Protocol

- Built a Python **multi-agent platform** with **MCP tool integration** and **Gemini ADK**, featuring a news agent that fetches live headlines on demand.
- Implemented **context engineering**, **LLM-as-judge evaluation**, observability, and security guardrails; deployed on **Vertex AI** and **Cloud Run** with **A2A** workflows.

Marksheet Management System [Demo](#)

Jul 2021 – Oct 2021

Flask · Bootstrap · MySQL · Pandas

- Built a **fully offline, single-machine** Flask app per the institution's requirement — reads student marks from **Excel sheets** with **Pandas** and applies **course-specific grading strategies** (PG/UG/Diploma).
- Generates print-ready **marksheets, exam hall tickets, and department copies** as PDFs and Excel reports — a stable build still running in production today.

Password Manager — Web & Desktop [Live: viswa2k.vercel.app](#) · [GitHub](#)

Sep 2020 – Dec 2020

Flask · Bootstrap · PyQt5 · SQLite · MongoDB · Firebase

- Built a password manager with **Flask + Bootstrap** (web) and **PyQt5** (desktop) clients, securing credentials with **Fernet symmetric encryption** (Python cryptography) and **hash-based login verification**.
- First end-to-end project — implemented persistence across **SQLite, MongoDB, and Firebase**, learning **CRUD design, encryption, and cross-platform GUI** development.

EDUCATION

MCA — Master of Computer Applications

Kongu Engineering College, Erode · Oct 2021 – Apr 2023

B.Sc Computer Science

Thiagarajar College, Madurai · Apr 2018 – May 2021

CERTIFICATES

RHCSA — Red Hat Certified System Administrator (EX200)

ID: 210-048-773 · 2021

Agent Development (Prototype to Production), LLM Architectures

Google Kaggle · 2025